

$P \equiv Q$ means $P \Leftrightarrow Q$ is always true

So:

- " P and Q are logically equivalent"
means
- " $P \Leftrightarrow Q$ " is a tautology.

Example:

$$P \Rightarrow Q \equiv \neg P \vee Q$$

because

$$(P \Rightarrow Q) \Leftrightarrow (\neg P \vee Q)$$

is always true.

Proof by Cases

The last type of proof that relies on direct proof is proof by cases. Suppose that we want to prove $p \Rightarrow q$ and this is difficult for some reason. We notice, however, that p can be broken into cases. Namely, there exist $p_0, p_1, \dots, p_{n-1}$ such that

$$p \Leftrightarrow p_0 \vee p_1 \vee \dots \vee p_{n-1}.$$

If we can prove $p_i \Rightarrow q$ for each i , we have proved $p \Rightarrow q$. If $n = 2$, then $p \Leftrightarrow p_0 \vee p_1$, and the justification of this is as follows:

$$\begin{aligned} (p_0 \Rightarrow q) \wedge (p_1 \Rightarrow q) &\Leftrightarrow (\neg p_0 \vee q) \wedge (\neg p_1 \vee q) \\ &\Leftrightarrow (q \vee \neg p_0) \wedge (q \vee \neg p_1) \\ &\Leftrightarrow q \vee \neg p_0 \wedge \neg p_1 \\ &\Leftrightarrow \neg p_0 \wedge \neg p_1 \vee q \\ &\Leftrightarrow \neg(p_0 \vee p_1) \vee q \\ &\Leftrightarrow p_0 \vee p_1 \Rightarrow q \\ &\Leftrightarrow p \Rightarrow q. \end{aligned}$$

Inference Rule (Proof by Cases)

For every positive integer n , if $p \Leftrightarrow p_0 \vee p_1 \vee \dots \vee p_{n-1}$ then

$$p_0 \Rightarrow q, p_1 \Rightarrow q, \dots, p_{n-1} \Rightarrow q \Leftrightarrow p \Rightarrow q.$$

A proof by contraposition or contrapositive proof

Indirect proofs

- Consider an implication: $p \Rightarrow q$
- It's contrapositive is $\sim q \Rightarrow \sim p$ is logically equivalent to the original implication.
- Thus, show that if $\sim q$ is true, then $\sim p$ is true .
- To perform an indirect proof, do a direct proof on the contrapositive.

Proof.

Assume $\sim Q$.

$\vdots$

Therefore, $\sim P$.

Thus, $\sim Q \Rightarrow \sim P$

Therefore, $P \Rightarrow Q$. ■

Ex 1: Suppose $x \in \mathbb{Z}$. If $7x + 9$ is even, then x is odd.

Ex 2: Suppose $x \in \mathbb{Z}$. If $x^2 - 6x + 5$ is even, then x is odd.

Ex 3: Suppose $x, y \in \mathbb{R}$. If $y^3 + yx^2 \leq x^3 + xy^2$, then $y \leq x$.

Ex 4: Suppose $x, y \in \mathbb{Z}$. If $5 \nmid xy$, then $5 \nmid x$ and $5 \nmid y$.

Ex5. Prove that if $n \geq ab$, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$,
where a and b are positive integers

Proof by contradiction

We now explore a third method of proof: **proof by contradiction**. This method is not limited to proving just conditional statements—it can be used to prove any kind of statement whatsoever. The basic idea is to assume that the statement we want to prove is *false*, and then show that this assumption leads to nonsense. We are then led to conclude that we were wrong to assume the statement was false, so the statement must be true. As an example, consider the following proposition and its proof.

Outline for Proof by Contradiction

Proposition P .

Proof. Suppose $\sim P$.

$\vdots$

Therefore $C \wedge \sim C$. ■

Example 1: Suppose $a \in \mathbb{Z}$. If a^2 is even, then a is even.

Ex. show that $\sqrt{2}$ is an irrational number.

Example 2: If $a, b \in \mathbb{Z}$, then $a^2 - 4b \neq 2$.

Example 3: If $a, b \in \mathbb{Z}$ and $a \geq 2$, then $a \nmid b$ or $a \nmid (b+1)$.

Ex. Prove that for all $x \in (0, \frac{\pi}{2})$, $\sin x + \cos x > 1$.

Note:

Proof of " $\forall x \in D, P(x) \Rightarrow Q(x)$ " by contradiction:

Let $\forall x \in D, P(x) \Rightarrow Q(x)$ -----(i)

Assume that (i) is **false**.

Then, $\sim(\forall x \in D, P(x) \Rightarrow Q(x))$.

i.e. $\exists x_0 \in D, P(x_0) \wedge \sim Q(x_0)$.

$\left. \begin{array}{l} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \right\} \begin{array}{l} \text{Definitions,} \\ \text{Axioms,} \\ \text{Logical equivalences,} \\ \text{Rules of inference} \end{array}$

..... is a **contradiction**.

Hence, (i) is **not false**. (i.e. Our assumption is **false**).

Thus, $\forall x \in D, P(x) \Rightarrow Q(x)$. ■

Ex1. for all integers n , if n^2 is odd, then n is odd

" $\forall n \in \mathbb{Z}, n^2 \text{ is odd} \Rightarrow n \text{ is odd}$ "

1. Universal Proof

1. Using predicate logic, prove: $\forall n \in \mathbb{Z}$, if n is even then n^2 is even.
2. Prove: $\forall n \in \mathbb{Z}$, $n(n + 1)$ is even.
3. Prove: $\forall x \in \mathbb{R}$, $x^2 \geq 0$.

2. Existential Proof

1. Prove: $\exists n \in \mathbb{Z}$ such that n is even and divisible by 6.
2. Prove: $\exists x \in \mathbb{R}$ such that $x^2 = 25$.
3. Prove: $\exists n \in \mathbb{Z}$ such that n is odd and n^2 is odd.

3. Direct Proof

1. Prove directly: $\forall n \in \mathbb{Z}$, if 4 divides n , then 2 divides n .
2. Prove directly: $\forall a, b \in \mathbb{Z}$, if a and b are odd then ab is odd.
3. Prove directly: $\forall x \in \mathbb{R}$, if $x > 2$ then $x^2 > 4$.

4. Proof by Cases

1. Using cases, prove: $\forall n \in \mathbb{Z}$, $n^2 + n$ is even.
2. Prove by cases that $|x| \geq 0$ for all real numbers x .
3. Using even/odd cases, prove: $\forall n \in \mathbb{Z}$, $n^2 - n$ is even.

5. Proof by Contraposition

1. Prove by contraposition: $\forall n \in \mathbb{Z}$, if n^2 is odd then n is odd.
2. Prove by contraposition: if ab is even, then a is even or b is even.
3. Prove by contraposition: if $x^2 \geq 9$, then $x \geq 3$ or $x \leq -3$.

6. Proof by Contradiction

1. Prove by contradiction that there is no integer that is both even and odd.
2. Prove by contradiction that $\sqrt{2}$ is irrational.
3. Prove by contradiction: there is no smallest positive rational number.

End of D10